



Stochastic Processes

Lecture 15

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DTMC on a Finite State Space

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Markov Chain)

A process $\mathbf{X} = \{X_n : n \in \mathbb{N}\}$ is called a **DTMC on a finite state space** $\mathcal{X}$ if for any $m, n \in \mathbb{N}$, times

$$s_1 < \cdots < s_m < t < t_1 < \cdots < t_n,$$

and values $(x_1, \dots, x_m) \in \mathcal{X}^m$, $(y_1, \dots, y_n) \in \mathcal{X}^n$, and $x \in \mathcal{X}$,

$$\begin{aligned} & \mathbb{P}\left(\underbrace{\left\{X_{s_1} = x_1, \dots, X_{s_m} = x_m\right\}}_{\text{before } t} \mid \underbrace{\left\{X_{t_1} = y_1, \dots, X_{t_n} = y_n\right\}}_{\text{after } t} \mid \left\{X_t = x\right\}\right) \\ &= \mathbb{P}\left(\left\{X_{s_1} = x_1, \dots, X_{s_m} = x_m\right\} \mid \left\{X_t = x\right\}\right) \cdot \mathbb{P}\left(\left\{X_{t_1} = y_1, \dots, X_{t_n} = y_n\right\} \mid \left\{X_t = x\right\}\right). \end{aligned}$$

Consequences of Markov Property

For a DTMC $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ on a finite state space $\mathcal{X}$:

- For all $m, n \in \mathbb{N}$, $(x_1, \dots, x_m) \in \mathcal{X}^m$, $(y_1, \dots, y_n) \in \mathcal{X}^n$, and $x \in \mathcal{X}$,

$$\begin{aligned} \mathbb{P}\left(\underbrace{\left\{X_{t_1} = y_1, \dots, X_{t_n} = y_n\right\}}_{\text{after } t} \middle| \left\{X_t = x, \underbrace{X_{s_1} = x_1, \dots, X_{s_m} = x_m}_{\text{before } t}\right\}\right) \\ = \mathbb{P}\left(\left\{X_{t_1} = y_1, \dots, X_{t_n} = y_n\right\} \middle| \left\{X_t = x\right\}\right). \end{aligned}$$

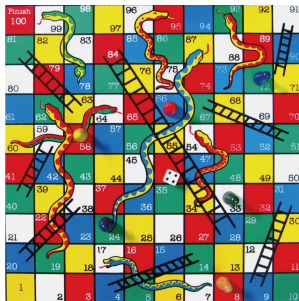
- For any $n \in \mathbb{N}$, $(x_1, \dots, x_n) \in \mathcal{X}^n$, and $x \in \mathcal{X}$,

$$\mathbb{P}\left(\left\{X_{n+1} = x\right\} \middle| \left\{X_1 = x_1, \dots, X_n = x_n\right\}\right) = \mathbb{P}\left(\left\{X_{n+1} = x\right\} \middle| \left\{X_n = x_n\right\}\right).$$

- FDDs:** For any $n \in \mathbb{N}$, $t_1 < t_2 < \dots < t_n$, and $(x_1, \dots, x_n) \in \mathcal{X}^n$,

$$\mathbb{P}\left(\left\{X_1 = x_1, \dots, X_n = x_n\right\}\right) = \mathbb{P}(\{X_1 = x_1\}) \cdot \prod_{t=2}^n \mathbb{P}\left(\left\{X_t = x_t\right\} \middle| \left\{X_{t-1} = x_{t-1}\right\}\right).$$

Example



- $\mathcal{X} = \{1, \dots, 100\}$
- Suppose $X_n = 6$ at some time n
- Then, $X_{n+1} \in \{7, 8, 31, 10, 11, 12\}$
- For all $x, x_1, \dots, x_{n-1} \in \mathcal{X}$, we have

$$\mathbb{P}\left(\left\{X_{n+1} = x\right\} \middle| \left\{X_n = 6, \underbrace{X_{n-1} = x_{n-1}, \dots, X_1 = x_1}_{\text{trajectory before } n}\right\}\right)$$

$$= \mathbb{P}\left(\left\{X_{n+1} = x\right\} \middle| \left\{X_n = 6\right\}\right).$$

Example: Random Walk

Let $\{X_n\}_{n \in \mathbb{N}}$ be an $\mathbb{N}$ -valued i.i.d. process. For each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^n X_i$.

- For any $n \in \mathbb{N}$ and $s_1 < s_2 < \dots < s_n < s_{n+1}$,

$$\begin{aligned}\mathbb{P}\left(\left\{S_{n+1} = s_{n+1}\right\} \middle| \left\{S_1 = s_1, \dots, S_n = s_n\right\}\right) &= \mathbb{P}\left(\left\{X_{n+1} = s_{n+1} - s_n\right\} \middle| \left\{S_1 = s_1, \dots, S_n = s_n\right\}\right) \\ &= \mathbb{P}\left(\left\{X_{n+1} = s_{n+1} - s_n\right\}\right) \\ &= \mathbb{P}\left(\left\{S_{n+1} = s_{n+1}\right\} \middle| \left\{S_n = s_n\right\}\right)\end{aligned}$$

- Thus, the process $\{S_n\}_{n \in \mathbb{N}}$ is a DTMC with state space $\mathcal{S} = \mathbb{N}$

Transition Probability Matrix (TPM)

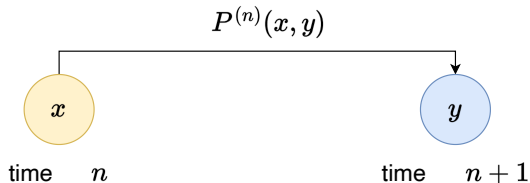
Transition Probability Matrix

Definition (Transition Probability Matrix)

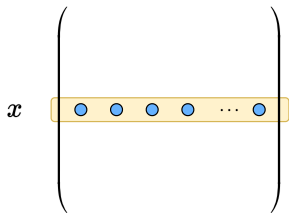
Let $\{X_n\}_{n \in \mathbb{N}}$ be a DTMC on a finite state space $\mathcal{X}$.

The **transition probability matrix (TPM)** of the Markov chain at any fixed time $n \in \mathbb{N}$ is a matrix $P^{(n)} = [P^{(n)}(x, y)]_{x, y \in \mathcal{X}}$ defined as

$$P^{(n)}(x, y) = \mathbb{P}\left(\left\{X_{n+1} = y\right\} \middle| \left\{X_n = x\right\}\right), \quad x, y \in \mathcal{X}.$$



Transition Probability Matrix



- For each $x \in \mathcal{X}$, $\sum_{y \in \mathcal{X}} P^{(n)}(x, y) = 1$.
- A matrix with non-negative entries and row sums equal to 1 is called a **row stochastic matrix**
- $P^{(n)}$ is row stochastic for every $n \in \mathbb{N}$
- Each row of $P^{(n)} \equiv$ PMF on $\mathcal{X}$
- For instance, if $\mathcal{X} \subseteq \mathbb{R}$, then

$$\sum_{y \in \mathcal{X}} y^2 P^{(n)}(x, y) = \mathbb{E}[X_{n+1}^2 | \{X_n = x\}],$$

where $\mathbb{E}$ above is $P^{(n)}(x, \cdot)$

- $P^{(n)}$ has a right eigenvector with eigenvalue 1
- $P^{(n)} \cdot \mathbf{1} = \mathbf{1}$, where $\mathbf{1}$ is the all-ones vector

Time Homogeneous DTMC

Definition (Time Homogeneous DTMC)

A DTMC $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ on a finite state space $\mathcal{X}$ and having TPMs $\{P^{(n)}\}_{n \in \mathbb{N}}$ is called **time homogeneous** if

$$P^{(n)} = P^{(n+1)} \quad \forall n \in \mathbb{N}.$$

That is, $P^{(n)}$ is same for all n .

In this case, we simply write P to denote the common TPM.

Remarks:

- Suppose $X_1 \sim \mu$, where $\mu = (\mu(x) : x \in \mathcal{X})$ is a PMF on $\mathcal{X}$. Then,

$$\mathbb{P}\left(\left\{X_1 = x_1, \dots, X_n = x_n\right\}\right) = \mu(x_1) \cdot \prod_{t=2}^n P(x_{t-1}, x_t).$$

- For a time homogeneous DTMC, the **initial distribution** μ and the TPM P govern the FDDs

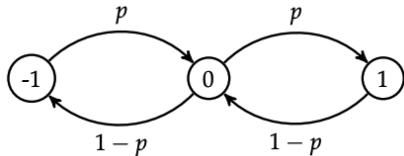
Shorthand representation: $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$

Transition Graph of a Time Homogeneous DTMC

Definition (Transition Graph)

Consider a time homogeneous DTMC $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ on a finite state space $\mathcal{X}$. The **transition graph** of the DTMC is a weighted, directed graph, say $G = (\mathcal{X}, \mathcal{E}, \mathbf{w})$, where:

- $\mathcal{X}$: node set of the graph.
- $\mathcal{E} = \{(x, y) \in \mathcal{X} \times \mathcal{X} : P(x, y) > 0\}$.
- $\mathbf{w} = \{w_{x,y} : (x, y) \in \mathcal{X} \times \mathcal{X}\}$, with $w_{x,y} = P(x, y)$ for all (x, y) .



$$\mathcal{X} = \{-1, 0, 1\}$$

$$P = \begin{pmatrix} 1-p & p & 0 \\ 1-p & 0 & p \\ 0 & 1-p & p \end{pmatrix}$$

Chapman-Kolmogorov Equation - 1

Theorem (Chapman-Kolmogorov, Part 1)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a timehomogeneous DTMC on a finite state space $\mathcal{X}$. For $k \in \mathbb{N} \cup \{0\}$, let

$$P_k(x, y) := \mathbb{P}\left(\left\{X_{k+1} = y\right\} \mid \left\{X_1 = x\right\}\right), \quad x, y \in \mathcal{X}.$$

Further, let $P_k = [P_k(x, y) : x, y \in \mathcal{X}]$. Then, the following hold.

1. $P_{k+\ell} = P_k \cdot P_\ell$ for all $k, \ell \in \mathbb{N} \cup \{0\}$.
2. $P_k = P^k = \underbrace{P \times \cdots \times P}_{k \text{ times}}$ for all $k \in \mathbb{N} \cup \{0\}$.

Proof of Theorem 1

- **Base case $k = 2$:**

For any $x, y \in \mathcal{X}$, we have

$$\begin{aligned}P_2(x, y) &= \mathbb{P}(\{X_3 = y\} \mid \{X_1 = x\}) \\&= \sum_{x' \in \mathcal{X}} \mathbb{P}(\{X_3 = y, X_2 = x'\} \mid \{X_1 = x\}) \\&= \sum_{x' \in \mathcal{X}} \mathbb{P}(\{X_2 = x'\} \mid \{X_1 = x\}) \cdot \mathbb{P}(\{X_3 = y\} \mid \{X_2 = x', X_1 = x\}) \\&= \sum_{x' \in \mathcal{X}} \mathbb{P}(\{X_2 = x'\} \mid \{X_1 = x\}) \cdot \mathbb{P}(\{X_3 = y\} \mid \{X_2 = x'\}) \\&= \sum_{x' \in \mathcal{X}} P(x, x') \cdot P(x', y) \\&= P^2(x, y).\end{aligned}$$

- The proof for $k > 2$ proceeds by mathematical induction (exercise)

Chapman-Kolmogorov Equation - 2

Theorem (Chapman-Kolmogorov, Part 2)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a time homogeneous DTMC on a finite state space $\mathcal{X}$. For each $n \in \mathbb{N}$, let $\mu_n = [\mu_n(x) : x \in \mathcal{X}]$ denote the PMF of X_n . Then,

$$\mu_{n+1} = \mu_n \cdot P \quad \forall n \in \mathbb{N}.$$

Proof of Theorem 2:

- For any $y \in \mathcal{X}$, we have

$$\begin{aligned} \mu_{n+1}(y) &= \mathbb{P}(\{X_{n+1} = y\}) \\ &= \sum_{x \in \mathcal{X}} \mathbb{P}(\{X_{n+1} = y, X_n = x\}) \\ &= \sum_{x \in \mathcal{X}} \mathbb{P}(\{X_n = x\}) \cdot \mathbb{P}(\{X_{n+1} = y\} \mid \{X_n = x\}) \\ &= \sum_{x \in \mathcal{X}} \mu_n(x) P(x, y). \end{aligned}$$

Random Mapping Representation

- Suppose that $X_1, X_2, \dots$ are IID, and let

$$S_n = \sum_{i=1}^n X_i = S_{n-1} + X_n$$

denote the position of the random walk at time n , with the convention $S_0 = 0$

- We saw earlier that $\{S_n\}_{n \geq 0}$ is a DTMC
- From the above relation, we have

$$S_n = f(S_{n-1}, X_n), \quad S_{n-1} \perp\!\!\!\perp X_n.$$

- **Is every Markov chain of this form?**

This forms the content of the **random mapping representation theorem**

Random Mapping Representation

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Random Mapping Representation)

Consider a TPM P on a discrete state space $\mathcal{X}$.

A **random mapping representation** of P is a tuple (Z, f) where

- $Z : \Omega \rightarrow \Lambda$ is a random variable,
- $f : \Omega \times \Lambda \rightarrow \mathcal{X}$ is a measurable function,

satisfying the condition

$$\mathbb{P}\left(\left\{f(x, Z) = y\right\}\right) = P(x, y) \quad \forall x, y \in \mathcal{X}.$$

Random Mapping Representation

Theorem (Random Mapping Representation)

Given a TPM P on a discrete state space $\mathcal{X}$, a random mapping representation always exists.

Proof of Theorem 3:

- The proof involves construction of $Z : \Omega \rightarrow \Lambda$ and $f : \Omega \times \Lambda \rightarrow \mathcal{X}$
- Let $\Lambda = (0, 1)$ and let $Z \sim \text{Uniform}(0, 1)$
- For each x , let $F(x, \cdot)$ denote the CDF corresponding to row x of P , i.e.,

$$F(x, \gamma) = \sum_{\substack{x' \in \mathcal{X}: \\ x' \leq \gamma}} P(x, x').$$

- Let $f : \Omega \times (0, 1) \rightarrow \mathcal{X}$ be defined as

$$f(x, z) = \inf\{\gamma \in \mathcal{X} : F(x, \gamma) \geq z\}.$$